1) Cancer Genomics Dataset

The cancer genomics dataset consists of gene-level features (columns named like gene_####), representing a high-dimensional feature vector per patient/sample. In my run, the feature matrix had 5479 gene features and 801 samples. Feature types include numeric gene measurements (each column corresponds to a gene feature), and labels are multi-class cancer types (e.g., BRCA, KIRC, etc.).

```
Successfully loaded heart disease data with 1025 rows
(1025, 13) {1: 526, 0: 499}
Duplicate rows before dedup: 723
Shape after dedup: (302, 13)
(801, 5479) {'BRCA': 300, 'KIRC': 146, 'LUAD': 141, 'PRAD': 136, 'COAD': 78}
```

I did two preprocessing steps: I dropped any gene columns with missing values, and ran experiments both with and without StandardScaler. In practice, scaling did not change cancer performance in my results (the metrics were identical with/without scaling, as we will later see). A few initial observations are that the classes are moderately imbalanced across 5 classes (largest class BRCA=300, smallest COAD=78). Not extreme, but imbalance can still bias a model toward the bigger classes. Also this dataset is extremely high-dimensional (5479 features) relative to sample size (801).

2) Model Performance (Heart Disease + Cancer Genomics)

Results for Heart dataset (scaler hidden, as identical):

```
Heart CV (no scaler)
  accuracy: mean=0.8045, std=0.0575
  precision_weighted: mean=0.8051, std=0.0579
  recall_weighted: mean=0.8045, std=0.0575
  f1_weighted: mean=0.8034, std=0.0582
  roc_auc_ovr_weighted: mean=0.8815, std=0.0479
```

Results for Cancer dataset:

```
Cancer CV (no scaler)
  accuracy: mean=0.9875, std=0.0124
  precision_weighted: mean=0.9880, std=0.0118
  recall_weighted: mean=0.9875, std=0.0124
  f1_weighted: mean=0.9874, std=0.0125
  roc_auc_ovr_weighted: mean=0.9999, std=0.0001
```

Much higher accuracy, precision, recall, and F1-score for cancer dataset over heart dataset. Empirically, cancer genomics was much easier for this model: CV accuracy ~0.99 and ROC-AUC ~1.0, compared to heart disease CV accuracy ~0.80. This suggests that, in this dataset, the gene expression patterns contain very strong signals that separate the cancer types.

The cancer dataset was much easier to classify. This difference might be from the nature of the feature spaces. The cancer dataset contains thousands of gene features, which may contain strong biological signals distinguishing cancer types. In contrast, the heart disease dataset has only 13 features, making it harder for the model to separate classes with the same degree of certainty.

Because the cancer dataset achieved extremely high performance (accuracy ≈ 0.98 and ROC-AUC ≈ 1.0), the number of classification errors was very small. This suggests that the model was able to identify strong patterns in the genomic data that differentiate the cancer types. In other words, the gradient boosting model was able to correctly identify the vast majority of samples in each class. The remaining errors likely occur in samples whose gene expression patterns are closer to another cancer subtype.

3) Hyperparameter Tuning (Gradient Boosting)

For the heart dataset, I tested depths {2,3,5}, estimators {50,100,200}, and learning rates {0.01,0.1,0.2}. For the cancer dataset, I used a smaller grid due to computational cost.

```
param_grid = {
    'max_depth': [2, 3, 5],
    'n_estimators': [50, 100, 200],
    'learning_rate': [0.01, 0.1, 0.2],
}
```

for heart, and

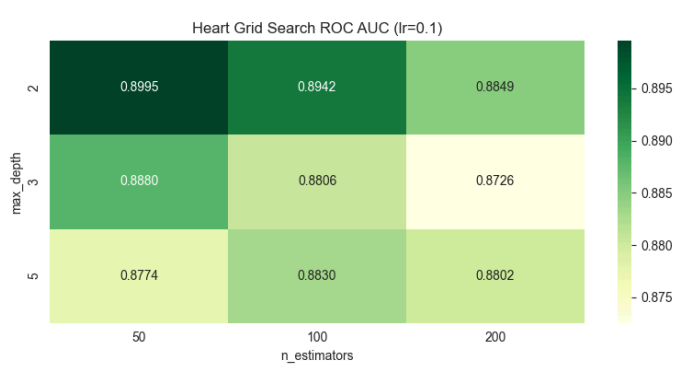
```
param_grid_c = {
    'max_depth': [2, 3],
    'n_estimators': [50, 100],
    'learning_rate': [0.05, 0.1],
}
```

for cancer.

Best parameters for the heart dataset were learning_rate=0.1, max_depth=2, n_estimators=50, with best accuracy of 0.868852.

accuracy	precision	recall	f1	roc_auc	max_depth	n_estimators
0.868852	0.818182	0.931034	0.870968	0.910560	1	50
0.836066	0.771429	0.931034	0.843750	0.891164	1	100
0.786885	0.735294	0.862069	0.793651	0.883621	1	200
0.786885	0.735294	0.862069	0.793651	0.845905	3	50
0.803279	0.757576	0.862069	0.806452	0.840517	3	100
0.819672	0.781250	0.862069	0.819672	0.836207	3	200
0.786885	0.750000	0.827586	0.786885	0.850216	5	50
0.770492	0.727273	0.827586	0.774194	0.837284	5	100
0.786885	0.750000	0.827586	0.786885	0.817888	5	200

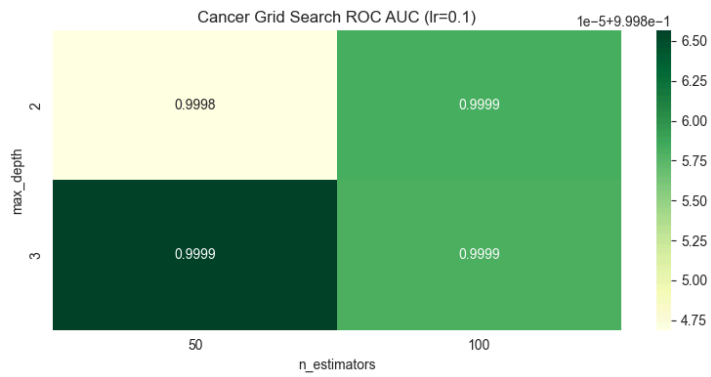
This performed well because heart disease has relatively few features and limited sample size, so shallower trees and fewer estimators likely help avoid overfitting while still capturing non-linear interactions.



For the cancer dataset, best parameters were learning_rate=0.1, max_depth=5, n_estimators=200, with best accuracy of 1.000000.

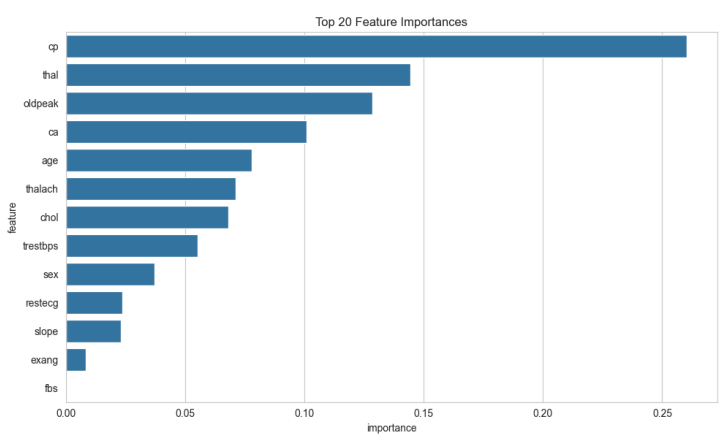
accuracy	precision	recall	f1	roc_auc	max_depth	n_estimators
0.987578	0.988885	0.987578	0.987701	1.000000	1	50
0.981366	0.984161	0.981366	0.981846	0.999906	1	100
0.981366	0.984161	0.981366	0.981846	0.998965	1	200
0.993789	0.993889	0.993789	0.993760	0.999891	3	50
0.981366	0.982774	0.981366	0.981229	1.000000	3	100
1.000000	1.000000	1.000000	1.000000	1.000000	3	200
0.981366	0.982774	0.981366	0.981229	0.999844	5	50
0.981366	0.982774	0.981366	0.981229	0.999953	5	100
1.000000	1.000000	1.000000	1.000000	1.000000	5	200

The cancer dataset contains very strong signals, allowing the model to separate classes easily. Increasing tree depth and number of estimators slightly improves performance but also increases the risk of overfitting.

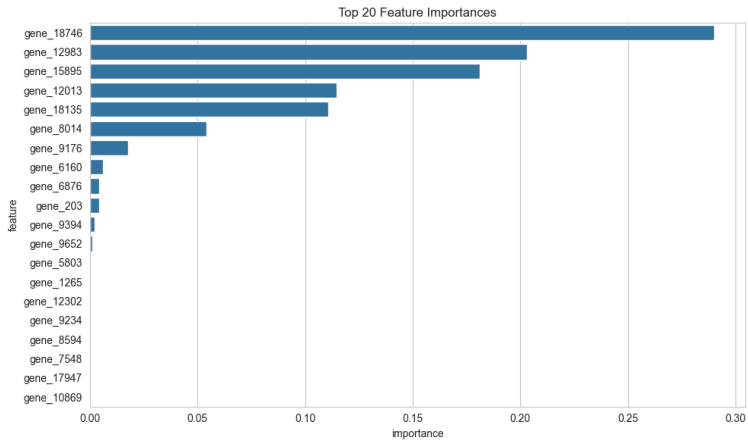


4) Feature Importance and Interpretation

Feature importance for heart dataset:



Top features include cp, thal, etc. Feature importance for cancer:



Top features include gene_18746, gene_12983, etc. Gradient boosting importance and Lasso "importance" measure different things. Lasso selects features that help a linear decision boundary, while gradient boosting importance measures how much a feature reduces impurity across tree splits (nonlinear, interaction-aware). So they may agree on a subset of "strong signal" genes, but they can also disagree because tree-based models can capture threshold effects and feature interactions while Lasso can miss nonlinear patterns but does better at global linear signal.

5) Comparison to HW1

I used gradient boosting feature importance to select top-K genes, then trained an HW1-style model on the full feature set, and reduced top-K feature sets for K ∈ {5, 10, 20}.

```
Full features (5479): acc = 0.9988 ± 0.0025
Top-5 GB features:    acc = 0.9900 ± 0.0030
Top-10 GB features:    acc = 0.9963 ± 0.0031
Top-20 GB features:    acc = 0.9988 ± 0.0025
```

Accuracy increases with more features, but the significant contributions clearly come from the top genes. Feature importance successfully identifies a small, highly informative subset of genes, demonstrating that the cancer types are driven by a handful of key genomic markers. From the data, the full feature and top-20 features perform identically, indicating that these genes do a good job at predicting the overall cancer diagnosis.

6) Discussion

Although gradient boosting models performed exceptionally well in their predictions, there are various limitations associated with these models. For instance, gradient boosting models are very sensitive to different hyperparameters such as depth of trees, number of estimators, and learning rate. When gradient boosting models increase their depth or number of estimators, they can easily overfit their models. Moreover, the high values of ROC-AUC scores for cancer genomics data, which are close to 1.0, could suggest that the model is detecting important biological signals. However, these high scores could also suggest that the model is overfitting to specific characteristics in the dataset. In particular, gradient boosting models could easily overfit when dealing with high-dimensional data in cancer genomics. Lastly, gradient boosting models could also perform poorly in clinical settings because clinical data is usually very noisy compared to data in a research setting.